Highlights

- An $Al_{13}Fe_4$ sphere inside an Al matrix is given a magnetostrictive strain.
- At 0 K the response is elastic; a dislocation above it shifts 0.8 Å and returns.
- A compliant interface layer weakens the atomistic field by about 20%.
- Above yield in a shell one radius thick at 40–60 MPa, a thin skin at 120–151 MPa.
- At $V^* = 70\,b^3$ the creep model needs 6–7 times the measured Fe–Al magnetostriction.